



SRI RAMAKRISHNA INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

20CS902 - FYP Phase II

SECOND REVIEW

BATCH NO : 39

ADDRESSING THE INCOME GAP CAUSED BY MIDDLEMEN IN AGRICULTURAL MARKETING THROUGH DIRECT FARMER-TO- CONSUMER SOLUTIONS

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CONTENT OUTLINE

- Progress Overview
- Challenges And Solutions
- Updated Methodology
- System Architecture
- Implementation Updates
- Results And Analysis
- Future Steps
- Conclusion
- References

PROGRESS OVERVIEW

- User registration and login system for farmers and consumers
- Product listing and management system
- Order placement and order tracking module
- Backend API development using Node.js and Express
- Database design and implementation using MongoDB
- Secure data handling and basic validation

CHALLENGES AND SOLUTIONS

Challenge 1: Integration Between Frontend and Backend

During development, connecting the frontend with backend APIs was difficult due to data handling and request errors.

Solution:

RESTful APIs were properly designed and tested using tools like Postman. Error handling and validation were improved to ensure smooth communication.

CHALLENGES AND SOLUTIONS

Challenge 2: Database Management

Managing large amounts of user, product, and order data efficiently was challenging.

Solution:

MongoDB collections and schemas were structured properly. Indexing and optimized queries were used to improve performance.

Challenge 3: Payment and Order Processing

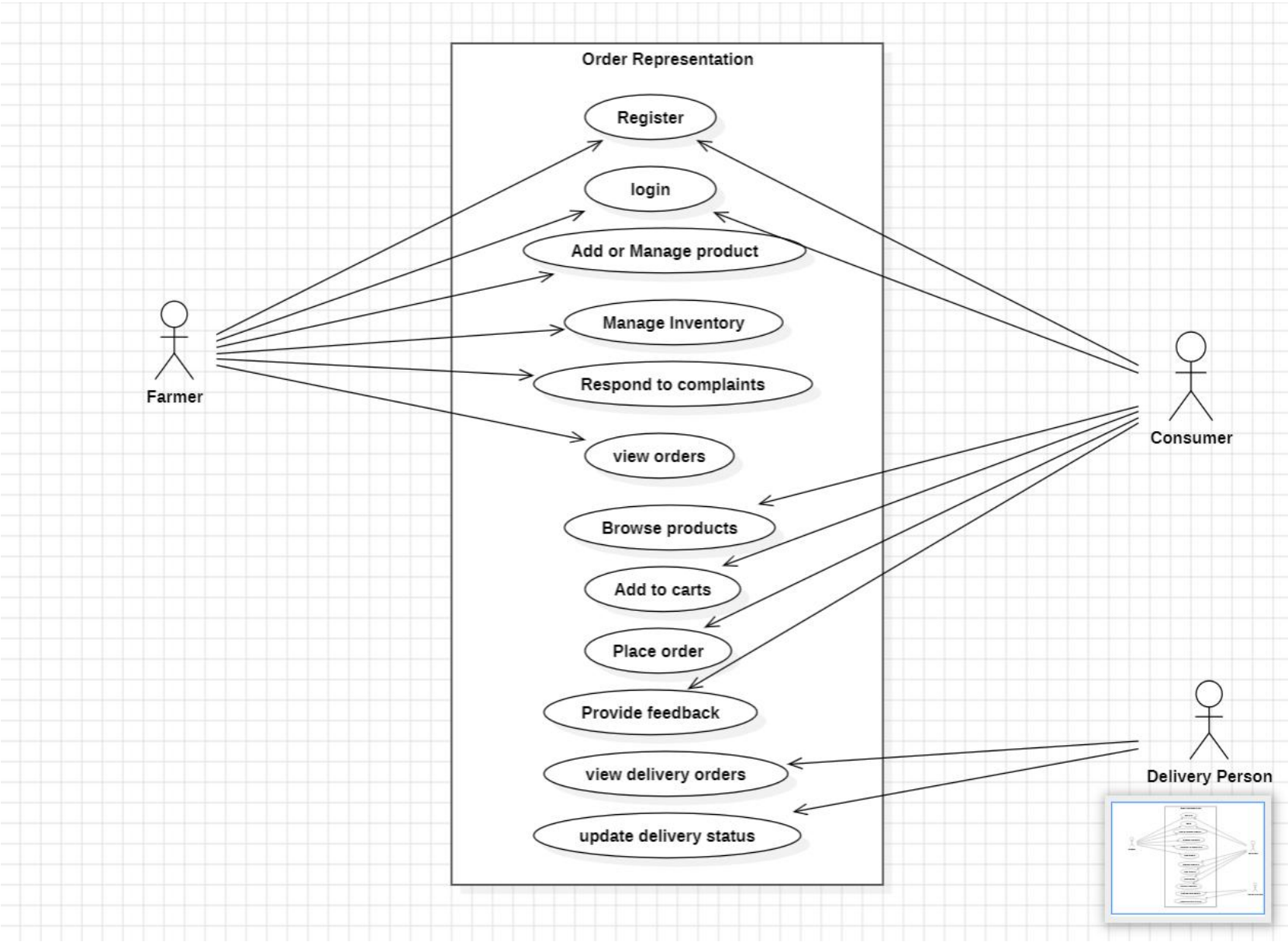
Handling order confirmation and payment verification required careful validation.

Solution:

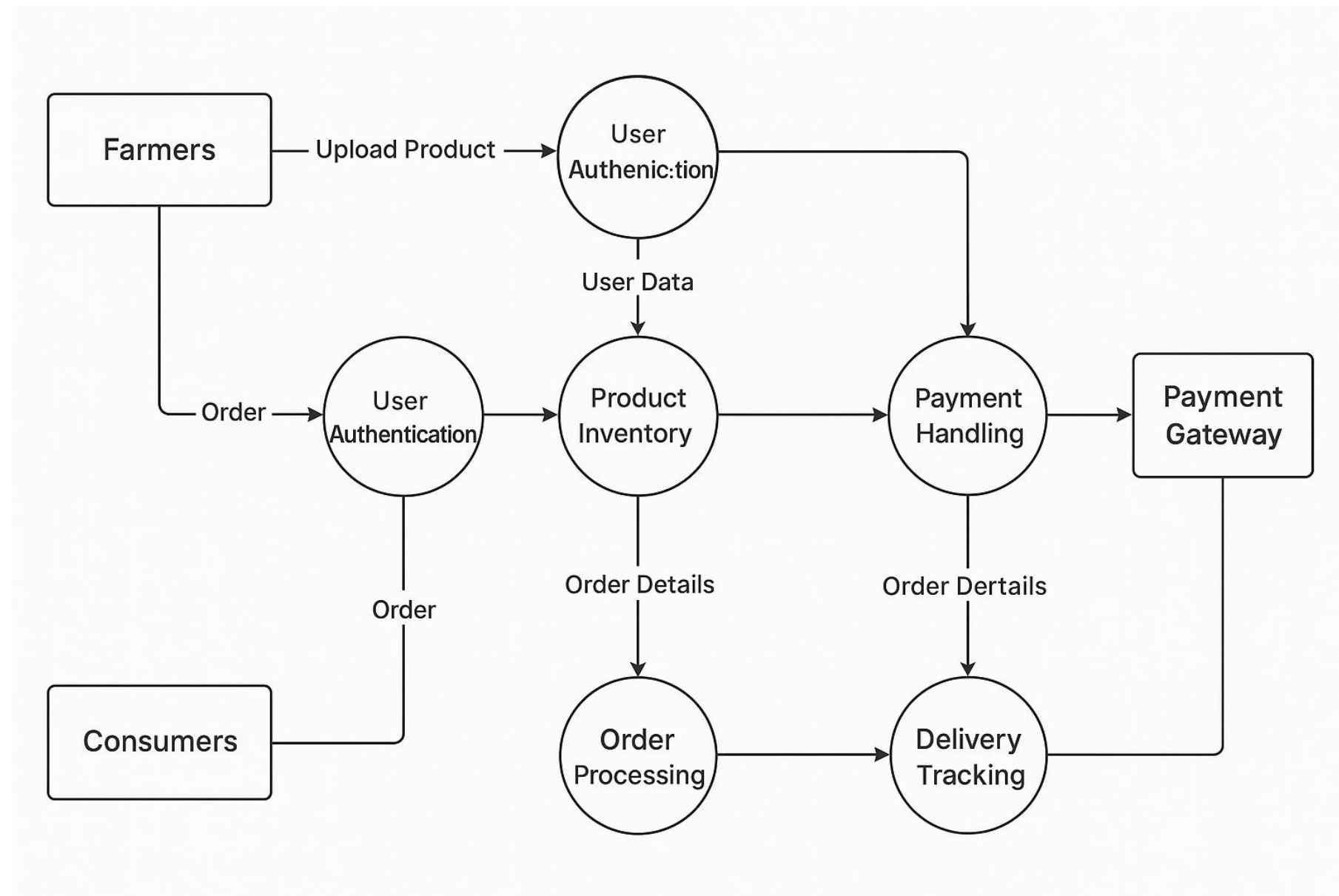
Backend validation and status tracking mechanisms were implemented to ensure reliable order processing.

UPDATED METHODOLOGY

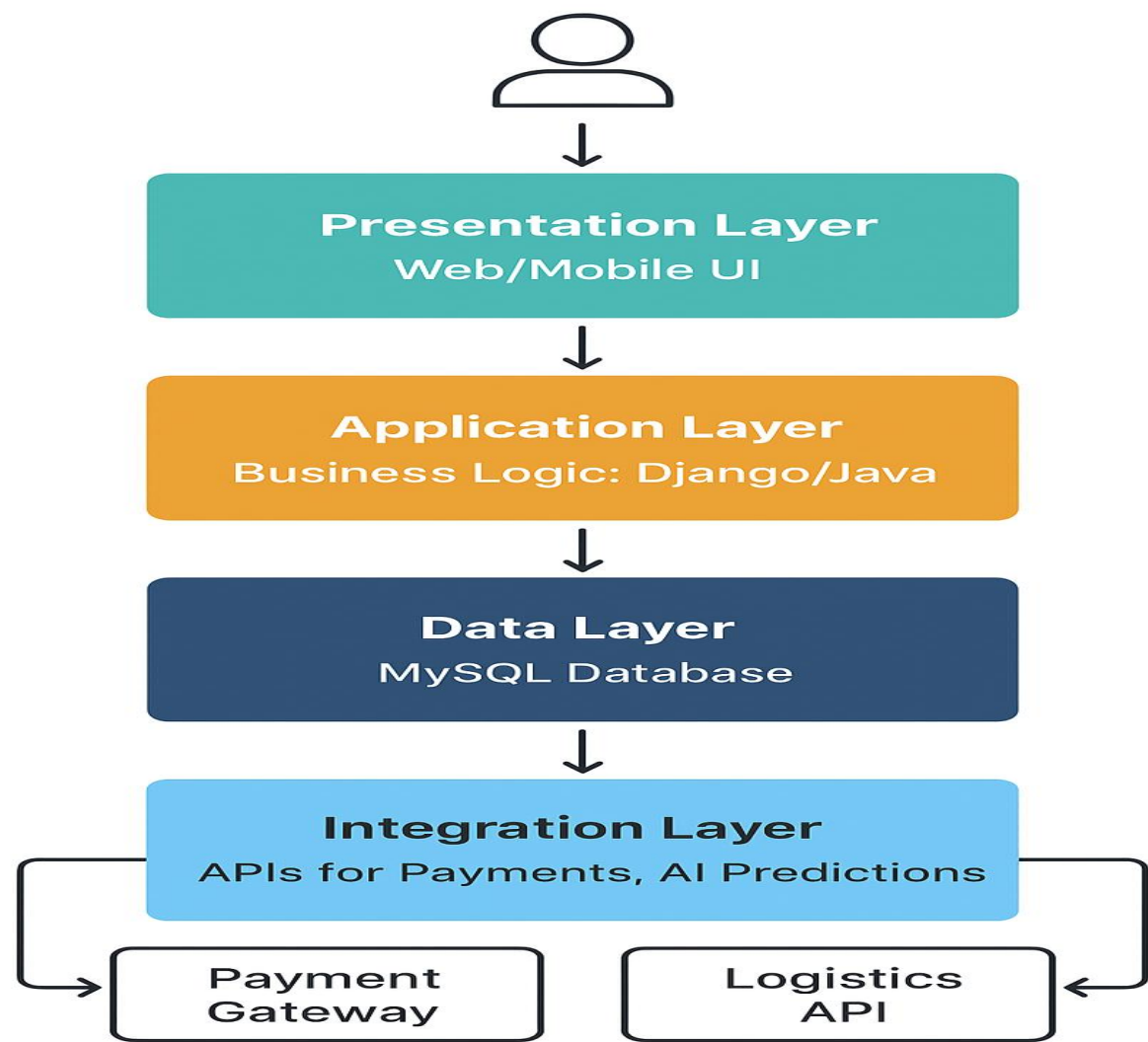
USE-CASE DIADRAM



DATA FLOW DIAGRAM



SYSTEM ARCHITECTURE



```
PS C:\Users\User\Documents\final year project\Farm-direct-connect-command> node server.js
Server started on port 5000
MongoDB Connected Successfully
```



Farm Direct Connect

Fresh Produce Directly from Farmers to Your Table – No Middlemen, Better Prices



Fresh Tomatoes

₹20/kg

📍 Location: Kerala

[View Details](#)



Fresh Bananas

₹60/dozen

📍 Location: Tamil Nadu

[View Details](#)

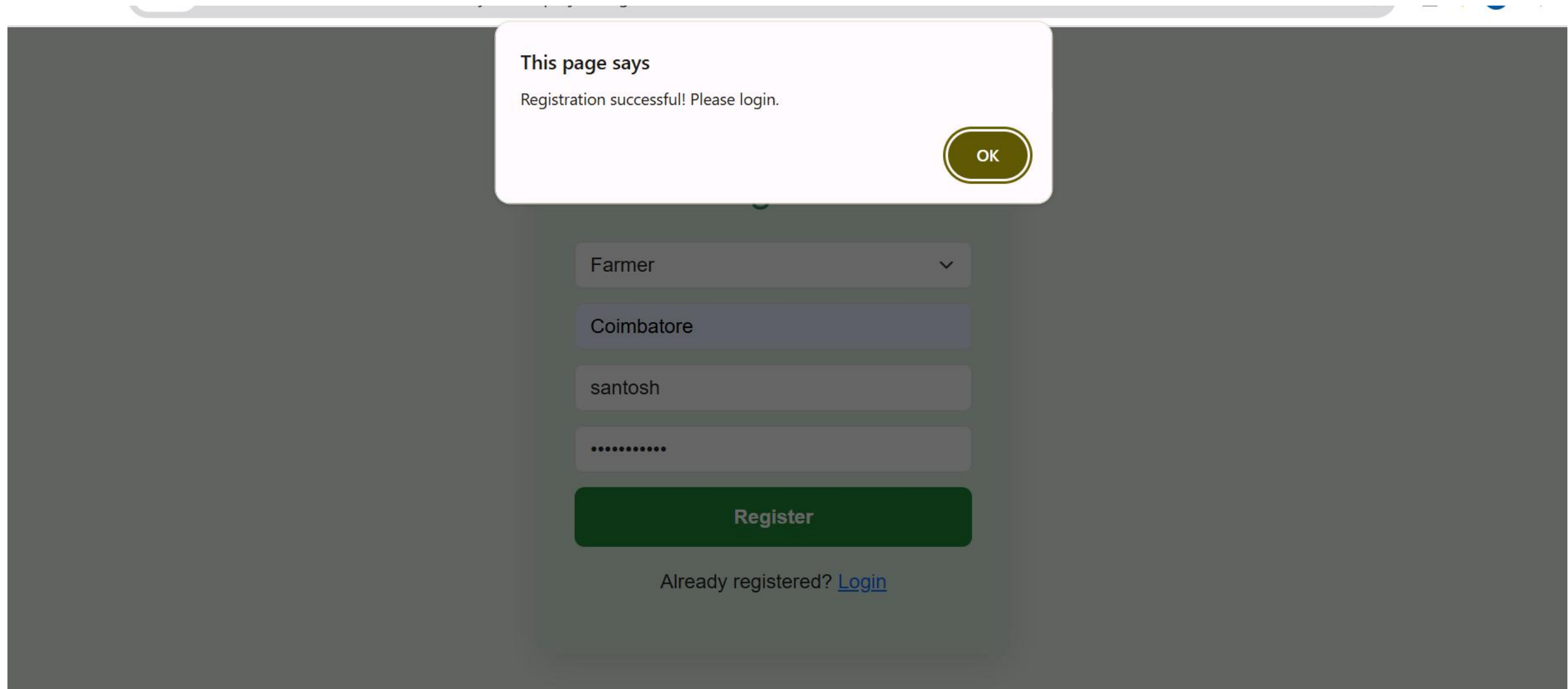


Basmati Rice

₹80/kg

📍 Location: Punjab

[View Details](#)



RESULTS AND ANALYSIS

- The platform successfully connects farmers and consumers directly.
- Farmers can upload and manage their products easily.
- Consumers can browse, place orders, and track deliveries.
- The system reduces dependency on middlemen.
- The project improves farmer income by reducing intermediary costs.
- Consumers get fresh products at lower prices.
- The platform ensures transparency in pricing and transactions.

FUTURE STEPS

AI Chatbot Integration

Implement an AI-based chatbot to assist users.

Provide 24/7 support for order status, product inquiries, and technical help.

Improve user experience through automated responses.

Delivery Partner Integration

Integrate third-party delivery services.

Enable real-time delivery tracking.

optimize delivery routes for faster service.

CONCLUSION

- The Direct Farmer-to-Consumer platform has been successfully developed using **HTML, CSS, JavaScript, Node.js, and MongoDB**. The system provides an efficient digital solution to reduce the income gap caused by middlemen in agricultural marketing.
- The project enables farmers to sell their products directly to consumers, ensuring fair pricing and better profits. Consumers benefit from fresh produce at affordable rates. The platform promotes transparency, trust, and sustainability.
- In conclusion, the project has achieved its objectives and provides a strong foundation for future enhancements and real-world deployment.

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THANK YOU